Configuring Apache Web Server

Session 4



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Objectives

- *Describe the process of configuring Apache Web Server*
- *Explain the use of Directives*
- *Explain the use of Modules*
- *Explain the process of modifying the default Index page*

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Apache Web Server: Configuration

- **`httpd.conf` file:**
  - Main configuration file
  - Located in `/usr/local/apache2/conf` directory
- **`./configure` command**
  - Used to configure from the source code
- **`–prefix=PREFIX` option**
  - Most important option in the configure command
  - Specifies the installation location of Apache Web server

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- Apache Web server is configured using different configuration options
- The basic configuration options are as follows:

Options	Description
`--configure-cache`	Configures the config.cache file
`--cache-file=FILE`	Records test results in the specified filename. This option is disabled by default
`--help`	Displays the help file
`--no-create`	Executes the configure script without generating the output files. This is used to verify the test results before generating makefiles for compilation
`--srcdir=DIR`	Defines the source file directory
`--version`	Displays the copyright information
`--quiet`	Does not generate messages during configuration (An alternative to this option is the --silent option.)

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Options	Description
`--enable-layout=LAYOUT`	Configures the source code and builds the scripts based on the installation tree specified in the LAYOUT. The config.layout specifies the different configuration. Apache server uses the default layout named LAYOUT
`--bindir=DIR`	Installs third party executables in the specified location
`--includedir=DIR`	Installs C header files in the specified location
`--information=DIR`	Installs information documentation to the specified location
`--libdir=DIR`	Installs object code libraries to the specified location
`--build=BUILD`	Defines the type of the system. For example, version number
`--disable=FEATURE`	Disables the specified feature
`--enable=FEATURE`	Enables the specified feature

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Multi Processing Modules

- Enable Apache server to run in different modes, to satisfy the requirements of every infrastructure
- Statically or dynamically included with the core
  - To include modules statically, the source code has to be added to the server source distribution
  - Dynamically included modules add functionality to the server by executing as shared libraries during start-up or restart of the server

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- Multi Processing Modules
  - mod_mine
  - mod_so
  - mod_log_config
  - mod_setevnif
  - mod_alias
  - mod_cgi
  - mod_userdir
  - mod_include
  - mod_dir
  - mod_status
  - worker
  - mod_vhost_alias
  - mod_rewrite
  - mod_expires
  - mod_proxy
  - mod_ssl

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mod_mime

- Allocates content metadata to the content by linking patterns or extensions in the filenames to the metadata values
- Associates the content's language, character set, and content-encoding to the file
- Allows to append a content-encoding entity-header field to a given file

Directive	Function
`AddCharset`	Specified content character set is mapped to the filename extensions
`AddEncoding`	Corresponding encoding type is mapped to the specified filename
`AddHandler`	Specified handler is mapped to the filename extensions
`AddLanguage`	Specified content language is mapped to the filename extensions
`AddType`	Specified content type is mapped with the filename extension
`TypesConfig`	Specifies the location of the `mime.types` file

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mod_so

- Loads modules and executable files into Apache server at runtime

Directive	Function
`LoadFile`	Links the named object files at startup or after a restart
`LoadModule`	Links the object file and includes the module structure to the list of existing modules

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mod_log_config

- Logs the client requests
- The log format can be customized and the log report can be generated and written to a file

Directive	Function
CookieLog	Specifies the filename used to log cookies
CustomLog	Defines the filename and the format of the log file
Logformat	Defines the log format or relates an explicit format with a nickname
TransferLog	Specifies the location of a log file

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mod_setevnif

- Enables to define internal environment variables
- The environment variables can be used by different modules of the server to decide the action to be executed

Directive	Function
`BrowserMatch`	Defines environment variables depending on HTTP User-Agent
`BrowserMatchNoCase`	Defines environment variables depending on User-Agent without respect to case
`SetEvnIf`	Defines environment variables on the basis of request attributes
`SetEvnIfNoCase`	Defines environment variables based on the basis of request attributes without respect to case

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- Allows to map the requests for URLs

Directive	Function
`Alias`	Links URLS to file system location
`AliasMatch`	Links URLS to file system location with the help of regular expression
`Redirect`	Transmits an external redirect (after seeking the client's consent) to obtain a different URL
`RedirectMatch`	Transmits an external redirect based on a regular expression match of the current URL
`RedirectPermanent`	Returns an external permanent redirect link. This directive instructs the client to request a different URL
`RedirectTemp`	Returns an external temporary redirect link. This directive instructs the client to request a different URL
`ScriptAlias`	Directs a URL to a file system location and defines the target as a CGI script
`ScriptAliasMatch`	Maps a URL to a file located on the Web server using a regular expression and defines the target as a CGI script

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- Enables execution of CGI scripts

Directive	Function
`ScriptLog`	Defines the location of the CGI script error log file
`ScriptLogBuffer`	Records the maximum number of PUT or POST requests in the scriptlog
`ScriptLogLength`	Defines the size limit of the CGI script log file

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mod_userdir

- The `mod_userdir` directive enables you to control access to specific directories. This module uses the `UserDir` directive

Syntax

```
UserDir directory-filename
```

where,

`directory-filename` - contains any one of the following options:

- Name of a directory
- The keyword disabled to stop username-to-directory translations
- The keyword disabled used with a list of usernames
- The keyword enabled used with a list of usernames

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mod_include

- Provides a filter to process request for files before sending them to the client

Directive	Function
SSIEnableAccess	Sets the -A flag when processing flow control based on conditions
SSIEndTag	Specifies the string (or character) to terminates an include element
SSIErrorMsg	Defines the error message to be displayed when an SSI error is encountered
SSIStartTag	Specifies the string (or character) to initiate an include element
SSITimeFormat	Defines the format to display date strings
SSIUndefinedEcho	Defines the string to display while a variable is not set and echoed
XHitBack	Parses SSI directives in files when the execute bit is enabled

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mod_dir

- Provides a trailing slash redirect, which helps in eliminating automatic index generation

Directive	Function
`DirectoryIndex`	Defines a list of resource to search when the user requests a directory
`DirectorySlash`	Enables or disables trailing slash redirects
`FallbackResource`	Sets a default URL for requests that do not contain a match

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mod_status

- Displays information related to the activities and performance of the server
- The following information is displayed in an HTML page:
  - Number of requests processed
  - Number of idle processes
  - Status of modules, number of requests served, and the total number of bytes processed
  - Server startup; and restart time
  - Averages of requests processed per second, and the number of bytes processed
  - Percentage of CPU used by Apache Web server and each module
  - Hosts and requests processed

Directive	Function
`ExtendedStatus`	Monitor extended status information for every request
`SeeRequestTail`	Verifies whether the first or the last 63 characters of the request is displayed when the number of characters in a request is greater than 63

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- Implements a hybrid server
- Uses multiple threads to process a number of requests
- Very stable and processes multiple requests with threads

Directive	Function
ThreadsPerChild	Defines the number of threads generated by each child process
MaxClients	Specifies the highest number of connections that will be processed at the same time
StartServers	Defines the number of child server processes to be created at startup
MinSpareThreads	Defines the minimum number of idle threads available to process requests
MaxSpareThreads	Defines the maximum number of idle threads available to process a request

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Directive	Function
`ServerLimit`	Defines the upper limit on the number of processes that can be configured
`ThreadLimit`	Defines the upper limit on the number of threads that can be configured per child process
`MaxRequestPerChild`	Defines the number of requests to be processed by an individual child server
`User`	Defines the user ID with which the server will process requests
`Groups`	Specifies the group for which the server will handle requests

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mod_vhost_alias

- Generates dynamically configured virtual hosts
- The IP address and the `Host: header` of an HTTP request help to determine the files to be served

Directive	Function
`VirtualDocumentRoot`	Defines the location of the document root for a specific virtual host
`VirtualDocumentRootIP`	Defines the location of the document root for a specific virtual host. This directive uses the IP address of the server as compared to the named virtual host.
`VirtualScriptAlias`	Defines the location of the CGI directory for a specific virtual host
`VirtualScriptAliasIP`	Defines the location of the CGI directory for a specific virtual host. It uses the IP address of the server to differentiate between multiple virtual hosts

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mod_rewrite

- Implements a rule-based rewriting engine to rewrite requested URLs
- Provides a flexible and strong URL manipulation mechanism

Directive	Function
RewriteBase	Defines the base URL for a directory rewrite
RewriteCond	Specifies a condition when the rewriting will take place
RewriteEngine	Enables or disables the rewriting engine
RewriteLock	Defines the name of the lock file used for managing the RewriteMap directive
RewriteLog	Defines the name of the file used for logging rewrite action it performs
RewriteLogLevel	Defines the verbosity of the log file used by the rewrite engine
RewriteMap	Defines a mapping function for searching a key
RewriteOptions	Defines options for the rewrite engine. The option can include inheritance of the parent server configuration
RewriteRule	Specifies rules for the rewriting engine

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mod_expires

- Manages the Expires HTTP header and the max-age directive of the Cache-Control HTTP header in a server response
- Expiry date can be relative to either the instance the source file was last modified or to the instance when the client last accessed the file
- These headers contain information about the validity of the file

Directive	Function
`ExpiresActive`	Enables creation of Expires headers
`ExpiresByType`	Sets the value of the Expires header configured by MIME type
`ExpiresDefault`	Defines the default algorithm for determining the expiry time

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mod_proxy

- Creates a gateway for the Apache Web server
- Incorporates a number of protocols and different load-balancing algorithms

Directive	Function
`ProxyRequests`	Enables redirecting of requests to the forward proxy server
`ProxyPass`	Maps the URL of remote servers into the local server URL space. This enables the remote servers to function as a proxy server
`<Proxy>`	Contains directives applicable to proxy servers
`ProxyBlock`	Contains resources that are prohibited from being proxied. The resources include words, host names or IP addresses, and domain names
`ProxyRemote`	Specifies the remote proxy to process specific requests
`NoProxy`	Specifies a list of subnets, hosts, and/or domains that will bypass the proxy and will be connected directly
`ProxyDomain`	Specifies the default domain name for proxied requests
`ProxyBadHeader`	Defines the method of processing bad header lines in a response

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mod_ssl

- Provides Secure Sockets Layer v2/v3 and Transport Layer Security v1 protocol for the Apache Web server

Directive	Function
`SSLCACertificateFile`	Defines the file consisting integrated PEM-encoded CA Certificates for client authorization
`SSLCARevocationPath`	Defines the directory where the Certificate Revocation Lists (CRL) of Certification Authorities (CAs) of clients are located to verify the communication
`SSLCertificateFile`	Specifies the PEM-encoded Certificate file for the Web server. In addition, it also directs to the corresponding RSA or DSA private key file

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Installing Modules in Bulk

- The configuration script in Apache server provides options that permit to enable or disable a number of modules at the same time
- The set of modules available in Apache server are categorized into three levels
  - Default
  - Most
  - All
- The modules included in the most category are also included in the default modules
- The modules in the All category are also included in the default and most modules

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Categorization of Modules

Default	Most	All	Explicitly Enabled
mod_access	mod_auth_anon	mod_cern_meta	mod_file_cache
mod_actions	mod_auth_dbm	mod_mime_magic	mod_charset_lite
mod_alias	mod_auth_digest	mod_unique_id	mod_cache
mod_asis	mod_dav	mod_usertrack	mod_disk_cache
mod_auth	mod_dav_fs		mod_mem_cache
mod_autoindex	mod_expires		mod_deflate
mod_cgi/mod_cgid	mod_headers		mod_ext_filter
mod_dir	mod_info		mod_proxy
mod_env	mod_rewrite		mod_proxy_connect
mod_http			mod_proxy_ftp

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Categorization of Modules

Default	Most	All	Explicitly Enabled
mod_imap			mod_proxy_http
mod_include			mod_ssl
mod_log_config			mod_suexec
mod_mime			
mod_negotiation			
mod_setenvif			
mod_so			
mod_status			
mod_userdir			

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Directives

- Instructions included in the configuration files that control the functioning of Apache Web server
- Those included in the `httpd.conf` file are applicable to the entire Web server
- Apache processes line-by-line in the `httpd.conf` file

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Allow Directive

- Provided by `mod_authz_host`
- Controls access of the hosts to a specific area of the server
- Access controlled by hostname, IP address, IP address range, or by other characteristics of the client request

Syntax

```
Allow from all|host|env=[!]env-variable [host|env=[!]env-variable] …
```

Snippet

```
Allow from example.com
```

The code enables access to the example.com domain

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Deny Directive

- Provided by `mod_authz_host` module
- Restricts access to the server based on the hostnames, IP address, or environment variable

Syntax

```
Deny from all|host|env=[!]env-variable [host|env=[!] env-variable] …
```

Snippet

```
Deny from 142.2.0
```

The code restricts access to the 142.2.0 user

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Order Directive

- Provided by the `mod_authz_host`
- Evaluates the order in the manner the Allow and Deny directives are evaluated

Syntax

```
Order ordering
```

Snippet

```
Order Deny Allow
Deny from all
Allow from example.com
```

The code allows access to hosts only from example.com

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Location Directive

- Limits the use of other enclosed directives by URL
- Processed in the order in which it appears in the configuration file

Syntax

```
<Location URL-path | URL> … </Location>
```

Snippet

```
<Location /status>
SetHandler server-status
Order deny,allow
Deny for all
Allow from 127.0.0.1
</Location>
```

The example displays a status report at the location 127.0.0.1

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<ifDefine> Directive

- Conditional directive
- Processed only if the test conditions specified at startup is fulfilled

Syntax

```
<IfDefine [!]parameter-name> … </Ifdefine>
```

Snippet

```
<IfDefine no_network>
<Location>
Order Deny,Allow
Deny from all
Allow from 127.0.0.1
   </Location>
</IfDefine>
```

The example enables network connectivity from 127.0.0.1 and denies for all others

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Include Directive

- Incorporates other configuration files from within the server configuration files at runtime

Syntax

```
Include file-path | directory-path
```

Snippet

```
Include /usr/local/apache2/conf/ssl.conf
```

The example specifies the directory in which the file should be included

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TypesConfig Directive

- Defines the location of the media types configuration file
- This file creates the default list of mappings from file extensions to content types

Syntax

```
TypesConfig file-path
```

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<IfModule> Directive

- Used to process the directives enclosed within it only if the module is present or absent
- Directives are processed only if the test returns true

Syntax

```
<IfModule[!]module-name…</IfModule>
```

Snippet

```
<IfModule core.c>
ServerName www.myserver.com: 80
</IfModule>
```

In the example, if the module core.c is included in Apache server, then the Server Name directive is processed

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Listen Directive

- Instructs Apache server to process requests from a specific IP addresses or port

Syntax

```
Listen[IP-address:]portnumber
```

Snippet

```
Listen 192.168.2.1:80
Listen 192.168.1.5:800
```

The example specifies that the Apache server should process requests from the specified port

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<Directory> Directive

- Contains a group of directives that can be applied to a specific directory or sub-directories
- Include a group of directives that will be applicable only to the named directory, sub-directories of that directory, and the files within the respective directories

Syntax

```
<Directory directory-path> … </Directory>
```

Snippet

```
<Directory ~ "^/var/ .*/[0-9]{3}">
```

The example would match directories in /var/ and that which consisted of three numbers

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<DirectoryMatch> Directive

- Encloses a group of directives that apply only to the named directory and to its subdirectories

Syntax

```
<DirectoryMatch regex> … </DirectoryMatch>
```

Snippet

```
<DirectoryMatch "/[A-Z][0-9](2)/">
```

The example matches directories in [A-Z] that consist of two numbers

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<Files> Directive

- Contains instructions that are specific to matched filenames

Syntax

```
<Files filename> … </Files>
```

Snippet

```
<Files~"\.gif.* ">
```

In the example, the Files directive matches all files with the extension .gif

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<FilesMatch> Directive

- Restricts the scope of the included directives by filename
- Same as the `<Files>` directive with the only difference that it accepts a regular expression

Syntax

```
<FileMatch regex> … </FileMatch>
```

Snippet

```
<FilesMatch "\.(gif|jpe?g)$">
```

In the example, FilesMatch directive is applied to all files with the extension of .gif and .jpeg

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<LocationMatch> Directive

- Includes instructions to be applied to regular expression matching URL
- Accepts a regular expression as an argument rather than a simple string

Syntax

```
<LocationMatch regex> … </LocationMatch>
```

Snippet

```
<LocationMatch"^/[A-Z][a-z]+/$ ">
```

The example enables to match the URLs containing words written in uppercase and lowercase

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SetHandler Directive

- Redirects all files with identical extensions when included in the `.htaccess` file or the `<Location>` or `<Directory>` section of the `httpd.conf` file

Syntax

```
SetHandler handler-name|None
```

Snippet

```
<Location>
        SetHandler server-information
</Location>
```

The example processes the requests that satisfy the conditions specified in the handler

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AccessFileName Directive

- Defines the name of the distributed configuration file
- Apache server matches the names of existing configuration file with the names from this list in every directory of the path to the document

Syntax

```
AccessFileName filename [filename] …
```

Snippet

```
AccessFileName .php
```

The example specifies that the server will search `/.php`, `/usr/.php`, `/usr/local/.php` and `/usr/local/web/.php` for directives before returning the document `/usr/local/web/index.html`

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AllowOverride Directive

- Contains the list of directives that are included in the `.htaccess` files
- If this directive is set to None, then the `.htaccess` file is ignored

Syntax

```
AllowOverride All|None|directive-type [directive-type]...
```

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Configure httpd.conf File

1. To change the directory, enter

```
cd /usr/local/apache2/conf
```

2. To copy the `httpd.conf` file, enter

```
cp httpd.conf httpd-cp.conf
```

3. To open the `httpd.conf` file using the vi editor, enter

```
vi httpd.conf
```

4. Press `I` to enter `INSERT` mode
5. To define the port, enter the following code in the `<Listen>` section of the `httpd.conf` file

```
Listen 192.168.0.220:80
Listen 127.0.0.1:80
```

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6. To set limitations for specified file extensions, enter the following code in the `<Order>` section of the `httpd.conf` file

```
<Files ~ "^\.ht">
Order Allow, Deny
Deny for all
</Files>
```

7. To define the location where the server information page is to be displayed, enter the following code in the `<Location>` section of the `httpd.conf` file

```
<Location /server-information>
SetHandler server-information
Order deny,allow
Deny for all
Allow from 192.168
</Location>
```

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8. Press Esc and `:wq!` to save changes to the `httpd.conf` file
9. Restart Apache Web server

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Index Page

- Is the first page that opens when accessing a particular site
- Contains links that enable to navigate through the site
- `DirectoryIndex` directive specifies the name of `index` file
- Module `mod_dir` provides the directive `DirectoryIndex`

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Modify Default index Page

1. Open the `gedit` text editor
2. Open the `index.html` file from `/usr/local/apache2/htdocs` directory
3. Delete the existing code and enter the code as shown

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Modify Default index Page


```
<html>
<head>
        <title>Custom Index Page</title>
</head>
<body>
<h1 align=”center”>MyApache.com</h1>
<hr>
<form>
<center>
        <h3>Search For:</h3>
        <input type=”text” name=”text1”>
        <input type=”submit” value=”Go”>
<br>
<a href=”/” align = “left”>Advanced Search</a>
    
<a href=”/” align=”right”> Preferences </a>
</center>
</form>
</body>
</html>
```

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4. Save the file
5. Open the Mozilla Firefox Web browser
6. Type `http://localhost/` in the address bar and press the Enter key



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Summary

- The main configuration file in Apache Web server is `httpd.conf`
- The `./configure` command compiles Apache from the source code
- The `--prefix` option of the configure command specifies the directory to install Apache server
- Directives are commands that set options and modules add functionality to the Web server
- Directory indexes are generated in the `index.html` file. Indexes can be generated using the `mod_autoindex` module

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